



Society of Petroleum Engineers

SPE-226598-MS

Physics-Informed Machine Learning for Hydraulic Fracturing—Part IV: Sustainable Fracturing with Carbon Score

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226598-MS](https://doi.org/10.2118/226598-MS)

This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

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Abstract

Hydraulic fracturing design has traditionally focused on production optimization; however, increasing environmental, social, and governance (ESG) pressures demand sustainability-driven workflows. This study extends previous physics-informed machine learning (PIML) work by incorporating carbon emissions as a secondary objective in a multi-objective optimization (MOO) framework. A synthetic physics-informed dataset was expanded to include emissions-related parameters. Proppant, water, chemical, and equipment associated carbon emissions were calculated using databases from GREET, EPA, LCA studies and other renowned models, combined with in-house models trained on real-time equipment metering. Random forest and XGBoost regressors were used to build predictive models for dimensionless productivity index (J_D) and total emissions (C) or dimensionless carbon score (C_{sd}), achieving R^2 of 0.7 and 0.96, respectively. Two optimization approaches were evaluated: (1) a coupled loop optimizing J_D and C simultaneously, and (2) a nested sequential optimization prioritizing production before emissions, where the prior was implemented due to its superiority. Pareto-front analysis showed the coupled approach captured superior trade-offs, enabling up to 15% emissions reduction with a minimal ~2% reduction in J_D simultaneously. Among the tested metaheuristics, the speed-constrained multi-objective particle swarm optimizer (SMPSO) converged faster and maintained better diversity than NSGA-II. This work demonstrates a robust methodology to jointly optimize productivity, economics, and emissions intensity in fracturing operations, providing a pathway toward sustainable completions. The workflow is extensible to field-scale real-time applications and can be adapted for basin-specific constraints, supporting ESG-driven decision-making while maintaining operational feasibility.

Introduction and Background

Hydraulic fracturing has revolutionized unconventional reservoir development by enabling the economic recovery of hydrocarbons from low-permeability formations. The design of hydraulic fracturing treatments has traditionally focused on maximizing production by optimizing fracture geometry and proppant placement. However, as the oil and gas industry transitions toward more sustainable practices, the environmental footprint of stimulation operations, particularly greenhouse gas (GHG) emissions, has

emerged as a critical design consideration. Regulatory frameworks, corporate environmental, social, and governance (ESG) commitments, and increasing public scrutiny demand that stimulation workflows evolve beyond production maximization to include emissions intensity as an explicit objective.

In the past decade, traditional supervised ML has been used by multiple researchers to optimize fracture design (He et al. 2019; Khan et al. 2020; Duplyakov et al. 2020; Khan et al. 2021a; Khan et al. 2022a), to generate production forecasts (Morozov et al. 2020), for fracture diagnostics and fluid measurements (Khan et al. 2021b; 2022b), rule based knowledge learning (Khan, 2024) and for greenhouse gas emissions prediction (Narang et al. 2023). But more recent advances in physics-informed machine learning (PIML), have opened new avenues for optimizing complex, multiphysics workflows (Karniadakis et al. 2021). Traditional data-driven ML approaches in stimulation design face well-documented limitations, including heavy reliance on large, high-quality datasets, poor generalization across basins, and limited interpretability. The PIML framework introduced in earlier parts of this work (Khan et al. 2024a; 2024b; 2025c) addressed these challenges by embedding physics-based constraints into ML models through a backbone trained on a synthetic dataset derived from classical fracturing theory. This backbone model successfully predicted key design parameters, fluid efficiency (FE), pad ratio (PR), proppant mass (M_{prop}), maximum proppant concentration (PPA_{final}), and dimensionless productivity index (J_D). The backbone model was then utilized with the transfer learning approach on two separate basins' real-field datasets and demonstrated improved generalization with similar accuracy. A downstream optimization loop, based on particle swarm optimization (PSO) and genetic algorithms (GA), further allowed the exploration of treatment designs that maximized J_D within an operationally constrained parameter space. The present study extends this framework to explicitly incorporate sustainability metrics by integrating carbon emissions as a complementary objective function. Hydraulic fracturing contributes to emissions through direct sources, such as diesel-powered pumping equipment, transportation, as well as indirect sources, including proppant manufacturing, chemical supply chains, etc. Life-cycle assessment (LCA) studies (Khan et al. 2025a) have shown that stimulation-related emissions can account for a significant proportion of a well's early-life carbon intensity, often determining whether a field meets ESG targets.

In this work, we propose a multi-objective optimization (MOO) framework that extends the existing synthetic backbone model to include the carbon emissions and couples the PIML-based productivity predictions with emissions life-cycle assessment to enable simultaneous production maximization and emissions minimization. This integration marks a paradigm shift and attempts to guide a discussion, from traditional single-objective fracturing design to a holistic, sustainable optimization framework. By leveraging physics-informed models, the proposed workflow ensures that the predicted outcomes remain physically consistent, interpretable, and transferable across basins, even with limited field data availability. The results presented here demonstrate that multi-objective optimization can achieve substantial reductions in carbon intensity with minimal impact on production, providing a path toward data-driven sustainable fracturing design.

The conceptual framework for this MOO workflow can be understood from Fig. 1 below. The analysis integrates various objective functions such as production performance (J or J_D), economic return (NPV), and carbon intensity (C or C_{sd}). The three panels represent different optimization scenarios as a function of fracturing treatment parameters (e.g., fluid volume, proppant mass, pump rate). The central idea is to analyze the derivatives $\frac{dy}{dV}$; where y is the objective function ($J_D, \text{NPV}, C_{\text{sd}}$) and V is the treatment volumetrics.

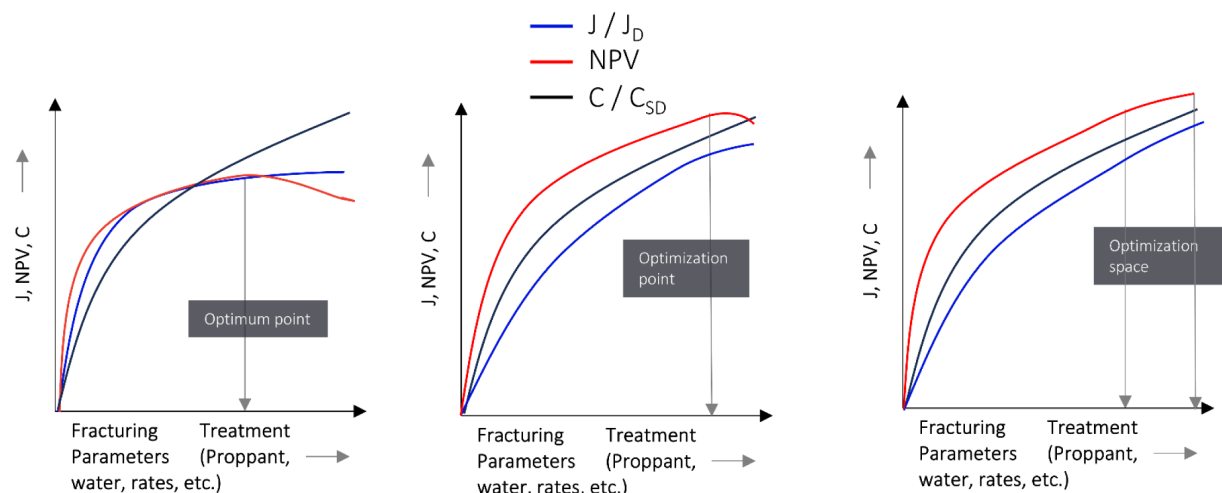


Figure 1—Conceptual representation of multi-objective optimization for hydraulic fracturing design. The optimization considers three competing objective functions: productivity, economics, emissions. Derivatives of the evolving objective functions are used to arrive to an optimum point or optimum space for the fracturing design.

In the first panel, the optimum point is determined by identifying the intersection of competing trends in productivity, economics, and sustainability. This represents a constrained optimization where the trade-offs are explicitly visible, and a single optimum design point is selected. Here the derivative approaches 0 for J_D , <0 for NPV, and stays constant and >0 for C_{SD} , hence the optimization solution is straight forward. The second panel shows a slightly different scenario in that the derivative for J_D *does not* approach 0 at the same rate as in scenario 1, rather shows a slight inflection, but still remains >0 . Here the economic return (NPV) dominates the optimization, resulting in a shifted optimum point compared to the first case. Here, higher NPV is achieved at the expense of a marginally higher carbon intensity, but still within acceptable operational constraints. The third panel demonstrates a more complex scenario where, even though inflections are observed in the objective functions, none of them approach 0 at a fast rate. Here, Pareto-based optimization space has to be considered with a range of optimal solutions exists instead of a single point.

Solution Approach

The first part of the solution was to build reliable emissions calculation to the existing synthetic fracturing dataset backbone. Next, to address the multi-objective nature of the problem with sustainability constraints, we initiated and evaluated two solution architectures:

Coupled Optimization Approach

The first workflow employs a fully coupled optimization loop, where both production and emissions outcomes are simultaneously evaluated during the design optimization process (Fig. 2). A machine learning model, trained on either a physics-based synthetic dataset or a curated real-field dataset, predicts production metrics and emissions intensity based on treatment inputs. The optimization loop varies the design/controllable parameters, such as pad ratio, proppant mass, maximum proppant concentration, water volume, and chemical totals, to search for the optimal set that jointly minimizes emissions while maximizing productivity. This multi-objective formulation is solved using a metaheuristic optimizer that identifies a trade-off surface without biasing one objective over the other unless explicitly weighted. If needed, the objective functions could be weighted by users to prioritize client needs. This approach captures the full spectrum of design trade-offs and ensures that no feasible low-emissions, high-productivity solutions are missed due to pre-optimization constraints. It is computationally more intensive but guarantees greater fidelity and interpretability for our solution approach.

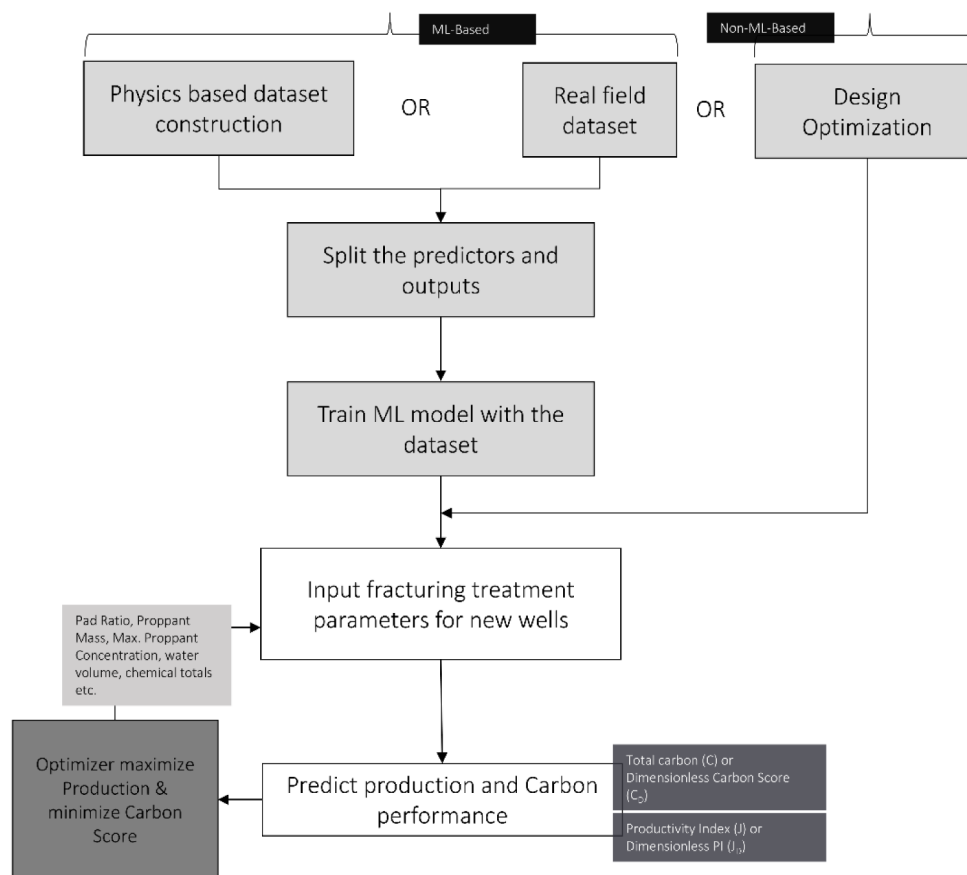


Figure 2—Conceptual schematic of the coupled optimization approach for fracturing design. A unified optimization loop jointly evaluates productivity and emissions intensity.

Nested (Sequential) Optimization Approach

The second solution structure involves a nested or sequential optimization logic (Fig. 3). In this architecture, the workflow first conducts a production-only optimization loop, identifying a region of interest in the design space based on maximum or near maximum productivity. Once the feasible high-productivity space is defined, a second loop is initiated to minimize the emissions metric within that constrained space. This strategy is computationally efficient and aligns with traditional decision-making workflows, where production targets are fixed and ESG impact is optimized subsequently.

However, the nested method risks missing Pareto-optimal solutions in cases where low-carbon, optimum (not maximum) production designs could outperform high-carbon, maximum-production options in overall project economics and regulatory compliance. Hence, while valuable for rapid screening, the nested approach may not fully capture the multi-objective balance required for our objective. Opposing objectives can contradict each other, potentially leading to a suboptimal solution. The second optimizer loop needs to be adjusted to define a new constrained search space within the first optimizer to try to find a solution that reduces emissions prioritizing the first optimizer or in other words prioritizing productivity.

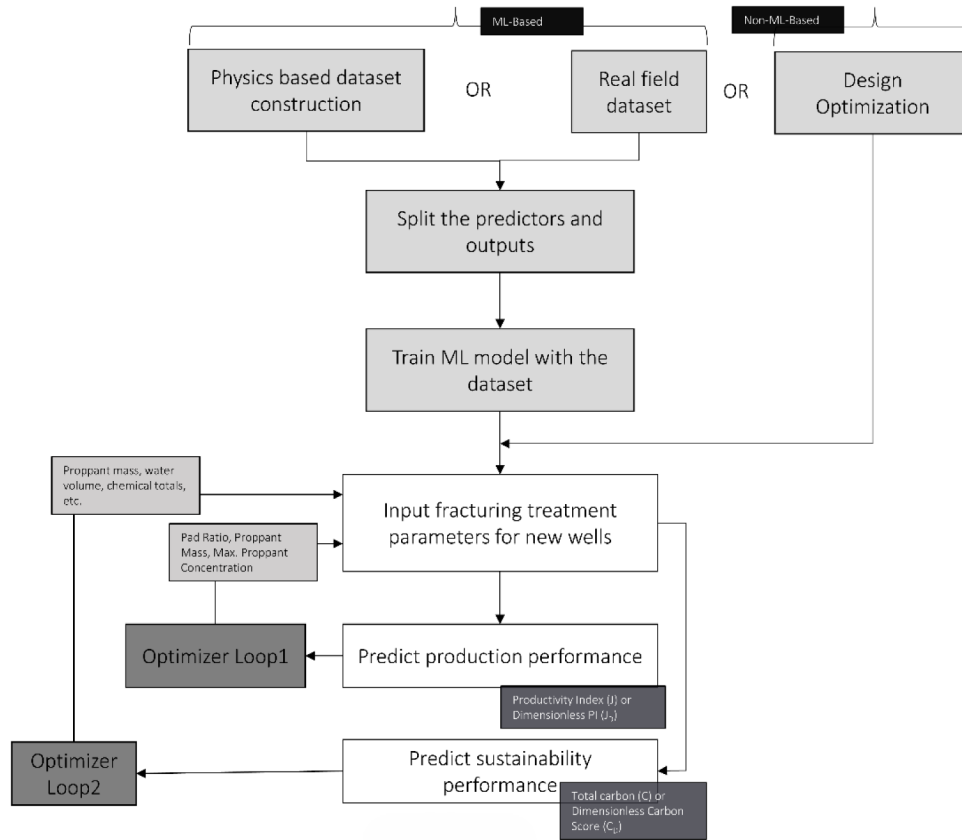


Figure 3—Schematic of the nested optimization approach with sequential loops. The first loop performs production optimization using an ML model to maximize productivity index. Once a high-performing treatment space is defined, a second loop searches for designs with minimum emissions within that region.

Methodology

The solution approach was implemented by extending the existing backbone of the synthetic dataset and conducting multivariate optimization along with predictive modeling. This section details the methodology for those steps:

Synthetic Dataset Extension

The central idea of synthetic dataset construction (detailed in the [Khan et al. 2024a](#)) is that we randomize a few independent variables, within practical ranges, and then calculate all the other parameters that have established some analytical or empirical dependence. This approach allowed us to create 10,000 rows of data for the study (but theoretically the dataset size has no limit). In this study we extended the same dataset to add the missing parameters for emissions quantification.

Normalized proppant factor. This factor calculated as $\text{CO}_2\text{e} / \text{proppant (tons/lb)}$ includes the manufacturing, supply chain, and last mile logistics associated with proppant ([equation 1](#)). We consider and randomize four types of proppants here, i.e., silica sand, ceramic, resin coated ceramic, and recycled ceramic. The embodied carbon and supply chain associated carbon is then randomized between ranges from various authenticated databases. Finally, the last mile transportation is calculated using a randomized distance (0-500 km) for wellsite from source.

$$p = p_e + p_{sc} + p_t \quad (1)$$

where, p_e is embodied carbon of proppant; p_{sc} is the total supply chain footprint for proppant in tons; p_t is the total last mile transportation footprint in tons.

Normalized water factor (w). This factor calculated as CO₂e / water (tons/gal) includes the last mile logistics associated with water mobilization from the pit to the wellsite. Similarly to the proppant, a randomized distance (0-500 km) for wellsite from water source is utilized.

Normalized chemicals factor. This factor calculated as CO₂e / clean fluid (tons/gal) includes the manufacturing, supply chain, and last mile logistics associated with chemicals (equation 2) and the components are the same as proppant but differs is how we derive the embodied and supply chain components. There are a few ways to approach these components, from simple to very complex. We evaluated a few options and the approach we converged upon was following – (a) randomize between two job types - conventional or unconventional basins; (b) create fluid packages for the two job types. For conventional, choose a holistic fluid package including gelling agent, crosslinker, breaker, surfactant, biocide, friction reducer, clay stabilizer. For unconventional, we only choose friction reducer and surfactant in the package; (c) consider the practical range of dosages for each of the products – minimum and maximum; (d) finally consider a range embodied carbon and supply chain associated carbon from various authenticated databases per unit mass of the product.

This provides us with a robust all-encompassing range of values for the carbon emissions associated with the chemicals.

$$m = m_e + m_{sc} + m_t \quad (2)$$

where, m_e is embodied carbon of chemicals; m_{sc} is the total supply chain footprint of chemicals in tons; m_t is the total last mile transportation footprint in tons.

Equipment factor. The final factor for equipment associated emissions (equation 3) calculated as CO₂e (tons) includes the emission from engine combustion during pumping (HP pumps, blender, hydration unit, etc.) and transportation logistics from the yard to wellsite. The components that go into this factor include the number of pumps which are in turn calculated by the HHP requirement of the treatment (equation 4). The injection rate range is randomized based on the conventional or unconventional basin and then backup HHP is added as a safety factor. The challenging elements of the calculation are the fuel efficiencies of each equipment. Here, the work presented by Khan et al. (2025b) was exceptionally useful. The authors there built learning from a regression model from the data captured by metering devices on various equipment.

$$e = (n \times e_{p_p} \times T_{pump}) + (n \times e_{p_i} \times T_{idle}) + ((e_{a1} + e_{a2} + \dots + e_{ai}) \times T_{pump}) + e_t \quad (3)$$

where,

$$n = \frac{BUF \times (WHP \times Q)}{40.8} \quad (4)$$

where, n is the number of HP pumps; WHP is the expected wellhead pressure, Q is the pumping rate, BUF is the backup factor which is 1.5 – 2.0 typically; e_{p_p} and e_{p_i} are the fuel efficiencies (gal/hr) for HP pumps when pumping and idling, respectively; e_{ai} is the fuel efficiency (gal/hr) for each auxiliary equipment such as blender, hydration unit, gensets.; e_t is the total transportation footprint in tons; T_{pump} and T_{idle} are the pumping and idling times.

Randomized range details. All the variables and parameters utilized for calculations, i.e., the randomization of parameters and the static values along with their reference has been detailed through multiple tables (Table A1 through Table A3) in Appendix-A.

Total carbon. The final calculation is for the absolute value of total CO₂e (tons) expressed in equation 5. It uses the above factors for a given design

$$C = (p \times M_{prop}) + (w \times V_{water}) + (m \times V_{CF}) + (e) \quad (5)$$

where, M_{prop} is the total proppant mass; V_{water} is the total water volume being brought to location; V_{CF} is the amount of water that was mixed with chemicals. Here note that M_{prop} and V_{CF} are outputs from the ML model built in the previous studies already, the same volumetrics are used to keep the dataset consistent. The; V_{water} is just a 10-20% additional water added for prime up, pressure test, dead volumes consideration.

Dimensionless carbon score. This is a dimensionless form of the carbon intensity expressed as carbon score. We calculated various forms of normalizations here

$$C_{\text{sd1}} = \frac{C_i - C_{\min}}{C_{\max} - C_{\min}} \left| \begin{matrix} 99 \\ 1 \end{matrix} \right. \quad (6)$$

$$C_{\text{sd2}} = \frac{C_i - C_{\min}}{C_{\max} - C_{\min}} \left| \begin{matrix} 95 \\ 5 \end{matrix} \right. \quad (7)$$

$$C_{\text{sd3}} = \frac{C_i - \mu}{\sigma} \quad (8)$$

$$C_{\text{sd4}} = \frac{\log(C_i)}{\log(C_{\max})} \quad (9)$$

where, C_{sd1} through C_{sd4} are the dimensionless carbon scores by (1) min-max normalization with values between 1 and 99 percentiles, (2) min-max normalization with values between 5 and 95 percentiles, (3) Z-score scaling, (4) log scaling, respectively; $C_i, C_{\min}, C_{\max}$ are the i^{th} , min, and max total carbon values; μ is the mean of range of C values, and σ is the standard deviation of range of C values.

Evaluation of C_{sd} approach. Since the synthetic dataset from reservoir data to geomechanical data all the way to fracturing, production, and emissions captures a large practical range of values, the distribution of the final dataset is right-tailed or positively skewed, as expected with the approach utilized. When we normalize the carbon value with various approaches, we can see this effect in the histogram of the C_{sd} values (Fig. 4). The log-scaled normalization visually seems better to represent the dataset, but it loses the linearly increasing trend utilized for the pareto analyses because the log-scaled value of a linear fit, trends logarithmically.

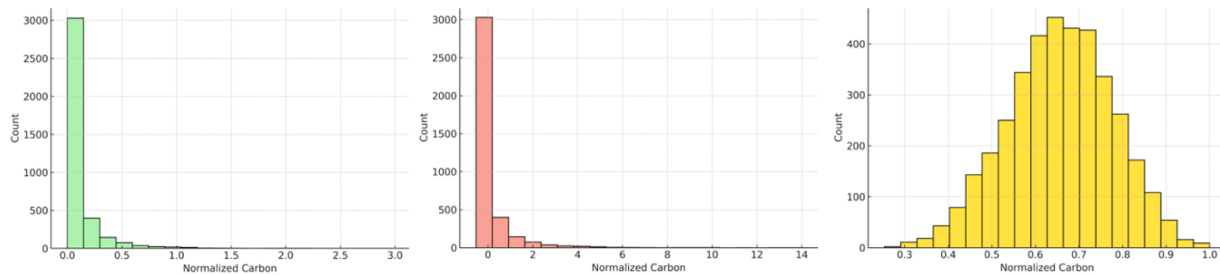


Figure 4—Histogram of the C_{sd} values with different normalization approaches: (left) C_{sd1} min-max normalization with P1-P99 distribution; (middle) C_{sd3} Z-score normalization; (right) C_{sd4} , log-scaled normalization.

Various approaches were utilized to clean the dataset to lower the scatter of the data outliers, but it was finally decided to capture the full dataset and variability. The trend lines with all the 5000+ data points show a very good linear fit ($R^2 = 0.67$) for the C_{sd} and a decent logarithmic fit ($R^2 = 0.27$) for J_D (Fig. 5). Various functions were fitted to the data but logarithmic function far exceeded the goodness compared to others. This clearly demonstrates the concept presented in figure 1, with actual calculations. Based on the goodness of the fit between all four approaches, C_{sd1} was finalized for the study.

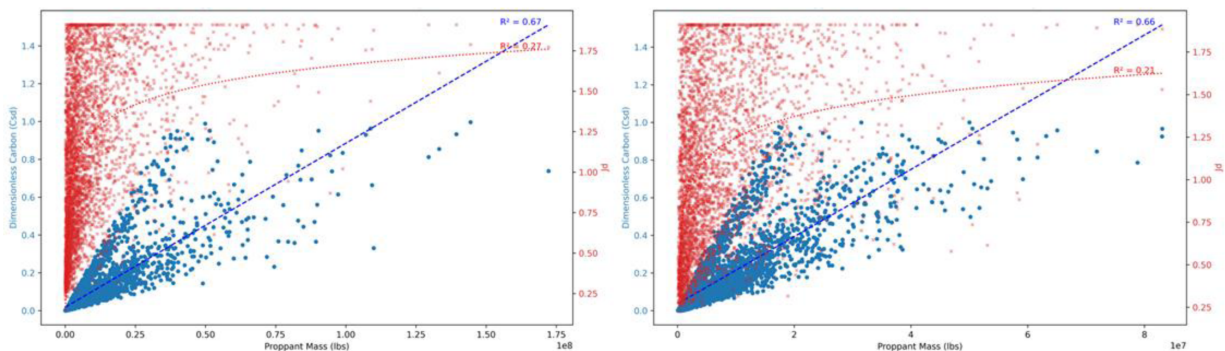


Figure 5—Scatter for J_D and C_{sd} evolution with increasing proppant mass showing linear increasing trend for C_{sd} and logarithmic increasing trend for J_D , aligned with the concept demonstration in figure 1: (left) C_{sd1} min-max normalization with P1-P99 distribution; (right) C_{sd2} min-max normalization with P5-P95 distribution.

Multivariable Optimizer

Multivariable optimization, also known as multi-objective optimization (MOO), is defined as the process of optimizing two or more conflicting objectives simultaneously within a given constrained search space. Unlike single-objective optimization, which focuses on finding a single most favorable solution, multivariable optimization seeks to find a trade-off solution that represents balance between the competing objectives. The solutions that find the trade-off balance generate the Pareto front; these are non-dominated, meaning that no single solution in the set is superior to another in all objectives (Deb 2011). In other words, improvements in one objective result would compromise the other.

In previous parts of this paper series (Khan et al. 2024a; 2024b; 2024c; 2025c) the focus has been on maximizing the dimensionless productivity index (J_D). However, in this study, the focus has extended to minimizing the expected emissions generated by the fracturing treatment. The objective is to find a balance between the ideal productivity index with minimal carbon emissions. The optimizer has to now consider multiple objectives in the optimization process when looking to the most favorable solution. To navigate the new complex optimization search space, two widely adopted algorithms were used and compared.

In this study the first employed optimization algorithm was Non-dominated Sorting Genetic Algorithm (NSGA-II). NSGA-II is a heuristic population-based algorithm inspired by the evolutionary natural selection (Yin 2003) to explore the search space. It uses a non-dominated sorting approach (Srinivas and Deb 1995) to classify the population into optimal solutions and introduces a crowding distance mechanism to preserve diversity among solutions (Deb 2002; Deb 2011). The algorithm operates by selecting, recombining, and mutating individuals in the population pool across generations, progressively refining the optimal solutions front with each iteration. The second employed algorithm was Speed-constrained Multi-objective Particle Swarm Optimization (SMPSO). This is a variation of the single objective Particle Swarm Optimization (PSO) optimization heuristic algorithm inspired by the social behavior of birds flocking or fish schooling (Kennedy and Eberhart 1995). This extension is for multi-objective optimization where the particles or potential solutions move through the search space influenced by the experience and behavior of their neighbors. SMPSO incorporates a velocity constraint that allows us to produce new effective particle positions in cases where the velocity becomes too high (Nebro et al. 2009). It also stores the non-dominated solutions found during the search and apply mutation operators to maintain diversity and avoid premature convergence. Both algorithms are well suited for nonlinear and complex search spaces.

The optimization process using either algorithm uses the different trained models to get the predictions. Each of the models has different model inputs, where the majority of the parameters are held constant as these are unchangeable representing uncontrolled factors. However, the fracture design parameters, that include fluid efficiency, pad ratio, total proppant mass, and maximum proppant concentration, are controllable design variables. Both models share these fracture design parameters are inputs and have a

unique importance for each model when generating the prediction. The search space for these variables was controlled by imposing multiple operational constraints that were varied for testing. The core idea behind this approach remains the same as in previous parts; once fracture design parameters are obtained, the optimizer can be used within the domain-specific constrained search space to find minor adjustments that can be made to the fracture design parameters. The minor adjustments could potentially lead to higher productivity with the lowest possible environmental impact. By integrating the multivariable optimizer with the predictive models, we can combine the strengths of both techniques when identifying the optimal fracture design parameters. This combined approach ensures the final solution aligns with practical, operational and now environmental considerations.

Results and Discussion

After completing the different preprocessing steps data were used for training and testing the different ML models to observe their performance and compare them to select the best model for the optimization. Three key evaluation metrics that included R^2 , root mean square error (RMSE), and mean absolute percent error (MAPE) were used to evaluate the performance of the models by comparing the predicted values against the actual values. Each metric provides unique insights to compare, assess, and evaluate the different trained models of this study. For instance, RMSE quantifies the average magnitude of the prediction errors expressed in the same units, making it interpretable but sensitive to outliers. MAPE, expressed as a percentage, makes the comparison easier and is also less sensitive to outliers. These metrics were combined with cross-validation to ensure a robust evaluation of model performance (Stone 1974). Cross-validation process produces an average performance measure across folds, helping to reduce bias and ensure that the model's performance is accurately reflected regardless of the training and testing folds. By incorporating multiple folds, cross-validation helps assess the model's generalizability, leading to a more robust, less error-prone model. For the comparative exercise though, mean absolute error (MAE) was used instead of MAPE as data was standardized for comparison. An average of the five-fold cross-validation was used to assess the model performance across the entire data and compare different ML models that included linear regression (LR), decision trees (DT), random forest (RF), XGboost (XGB), support vector regressor (SVR), k-nearest neighbors (KNN), and neural networks using a multilayer perceptron (NN). The comparison of the two models J_D and C can be observed in Fig. 6 using standardized values. It can be observed that RF and XGB were the two best performing models for both cases with highly similar results. Nevertheless, in most cases, RF average results slightly outperformed XGB.

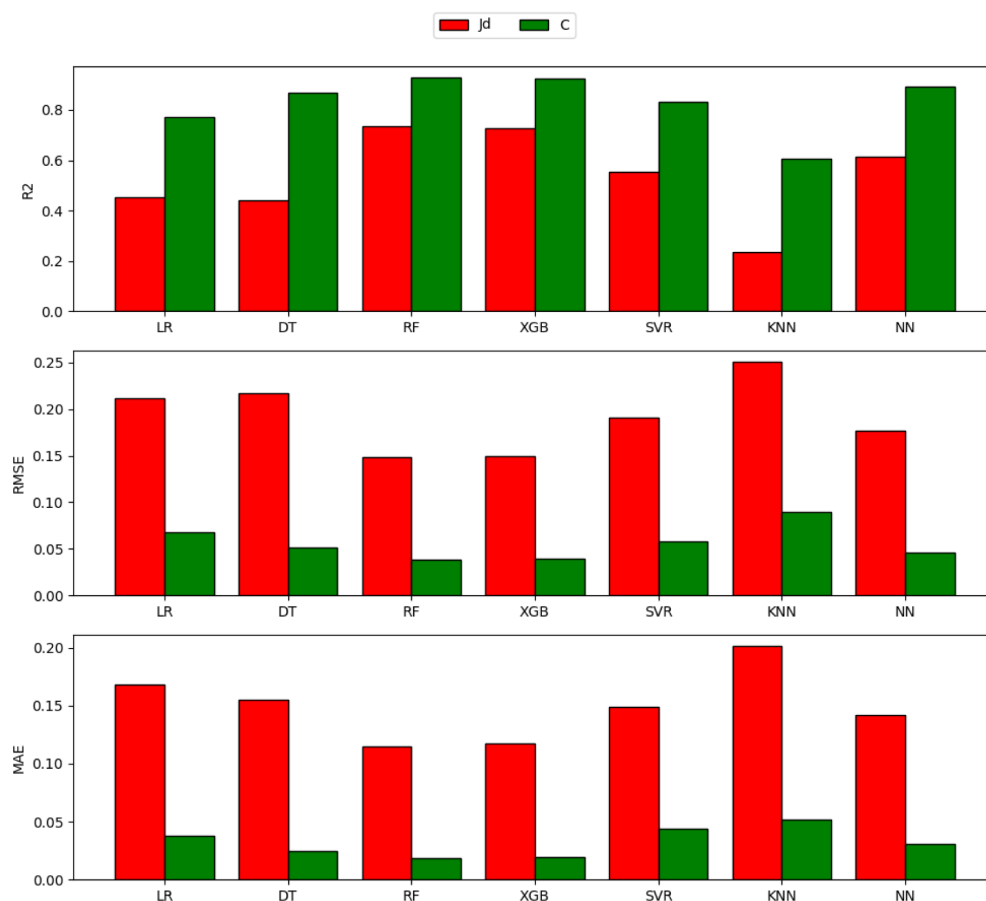


Figure 6—ML model comparison using LR, DT, RF, XGB, SVR, KNN, and NN for J_d and C. Values are standardized for comparison.

The predictive J_d and C models were trained and tested using only synthetic data. During cross-validation, the same test folds were used for both models to ensure consistency during the evaluation. No transfer learning (with real-data as in parts-II and III of the series) was done in this workflow to focus on the optimizer component with the proposed methodology. The J_d model results comparing different folds show an even performance regardless of what data subset was used for testing observed by looking at the similarity between folds across the different metrics (Fig. 7, top). The average results showed an R^2 of 0.7, RMSE of 0.07 and MAPE of 24.1%, exhibiting a good performance and predictability. However, by looking at the distribution of the scatter in the figure, it can be observed how the predicted model is overestimating the prediction for lower values and underestimating the predictions compared to the higher values. Predicting J_d is more complex compared to the model for execution emissions (C), which exhibited an excellent performance across folds (Fig. 7, bottom). The average results showed an R^2 of 0.967, RMSE of 6.62 equivalent MMCO₂e-tons, and MAPE of 12.4%. The predicted versus actual distributions are even throughout but the prediction become less reliable at higher values as values diverge from the unit slope line.

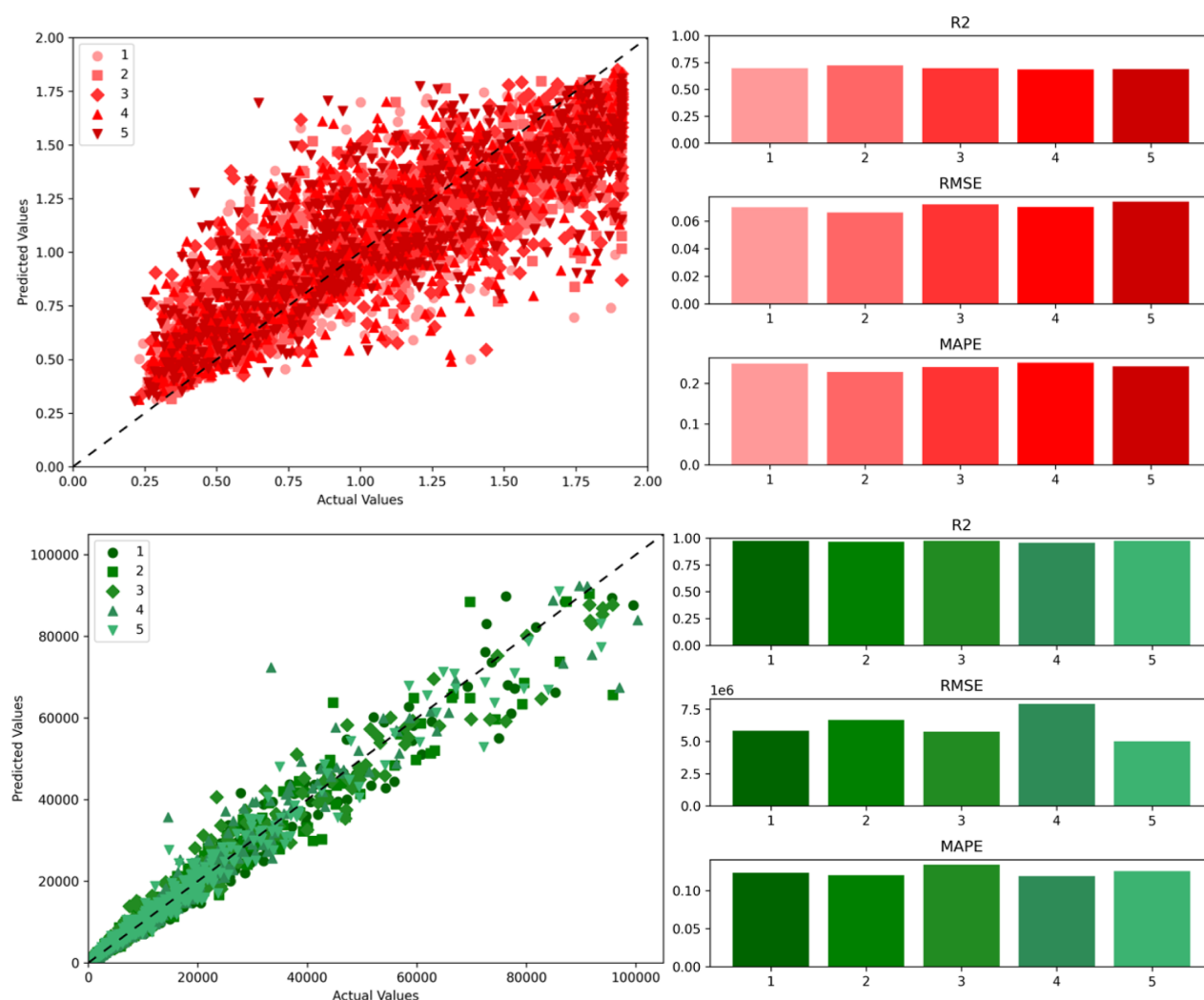


Figure 7—Results comparison showing the predicted against actual values along with the R², RMSE and MAPE evaluation metrics for all folds in the cross validation: (top) J_D; (bottom) C – total emissions.

As both models will be used in the optimization process, it is important to understand which variables are impacting the model performance because among all the predictors of both models only four are being optimized searching for a combination of fracture design parameter that would lead to an optimal value (higher J_D and lower C). Fig. 8 displays the top variables with the highest feature importance, dominating the predictions of the RF models for both J_D and C cases. The feature importance analysis revealed that among all predictors; proppant mass contributed the most to the predictive performance of both models. In the J_D model it can be observed how the pad ratio (PR_{Safe}) and fluid efficiency for crosslinked fluid (Eff-XL) were also among the top predictors with high feature importance. While the C model was highly dominated by the proppant mass along with ceramic proppant, volume of water, etc., with little importance to the other fracture design variables. This feature ranking shows explainable causality and gives confidence for further downstream modeling and optimization.

In model refinement, feature engineering techniques can be applied to improve the contribution of less influential predictors. For example, creating interaction with other variables, applying non-linear transformations, or other techniques can enhance the relevance to the model. Increasing the importance of certain predictors can potentially improve the model's performance by capturing additional variance and new complex patterns. Nevertheless, this process must be approached cautiously, as introducing engineered features can also increase the risk of overfitting, complexity, multi-collinearity that could

increase computational cost, reduce interpretability yielding little to no actual improvement in overall model performance.

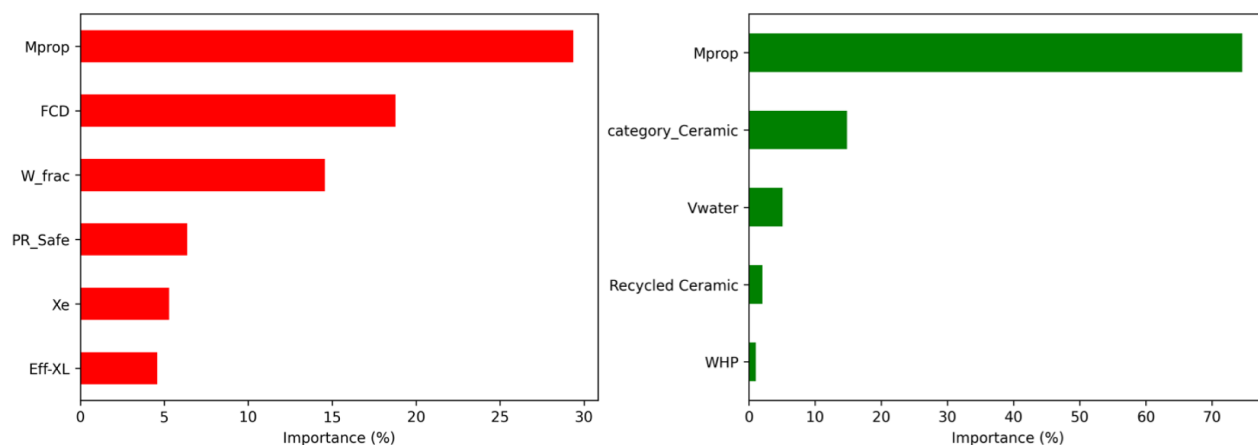


Figure 8—Feature importance analysis of trained models for JD (red) and C (green) showing the top predominant variables.

The last step of the proposed workflow and the main focus on this is incorporating an optimizer for the two critical objective functions to minimize emissions and maximize productivity. Both models are regressions where each take different predictors as inputs (as the dataset was extended for emissions with new variables added which are not relevant for J_D predictions) but share the fracture design parameters. These are the variables that are optimized to search for a combination that could potentially increase the J_D and minimize the CO_2e -tons associated with the fracture design variables. The search space is restricted by imposing constraints and applying penalties to steer the optimization process toward practical and operationally feasible solutions. By allowing the fracture design parameters to vary within a constrained range of $\pm 10\%$ around their baseline values, the optimizer was able to explore a diverse set of feasible solutions that balance the competing objectives. The optimal solutions build the pareto front with $\pm 10\%$ as illustrated in [Fig. 9, top](#) using both SMPSO and NSGA-II. The output of optimal solution in this case would be the combination at the front of the curve before it plateaus and the tradeoff becomes minimal. For instance, there is an insignificant reduction in emissions with a significant reduction in productivity.

If the percentage of varying freedom is expanded for the fracture design parameters, the model has more flexibility to further expand the pareto front. [Fig. 9, bottom](#) shows the same values with a constrained range of $\pm 20\%$ around their baseline fracture design values. The pareto front expands, leading to higher J_D and lower C , as expected. As such it is important to properly select the constraints based on feasible ranges. Overall, the results clearly show the trade-offs between increased production and reduced emissions, with each point on the curve representing a unique combination of input fracture design parameters that satisfies both models simultaneously. As expected, solutions with higher production are associated with higher emissions. As a result, in order to minimize the emissions a degree of production is sacrificed in the trade-off. This outcome highlights the value of the multivariable optimization approach in finding alternative designs that best align with operational priorities and environmental constraints.

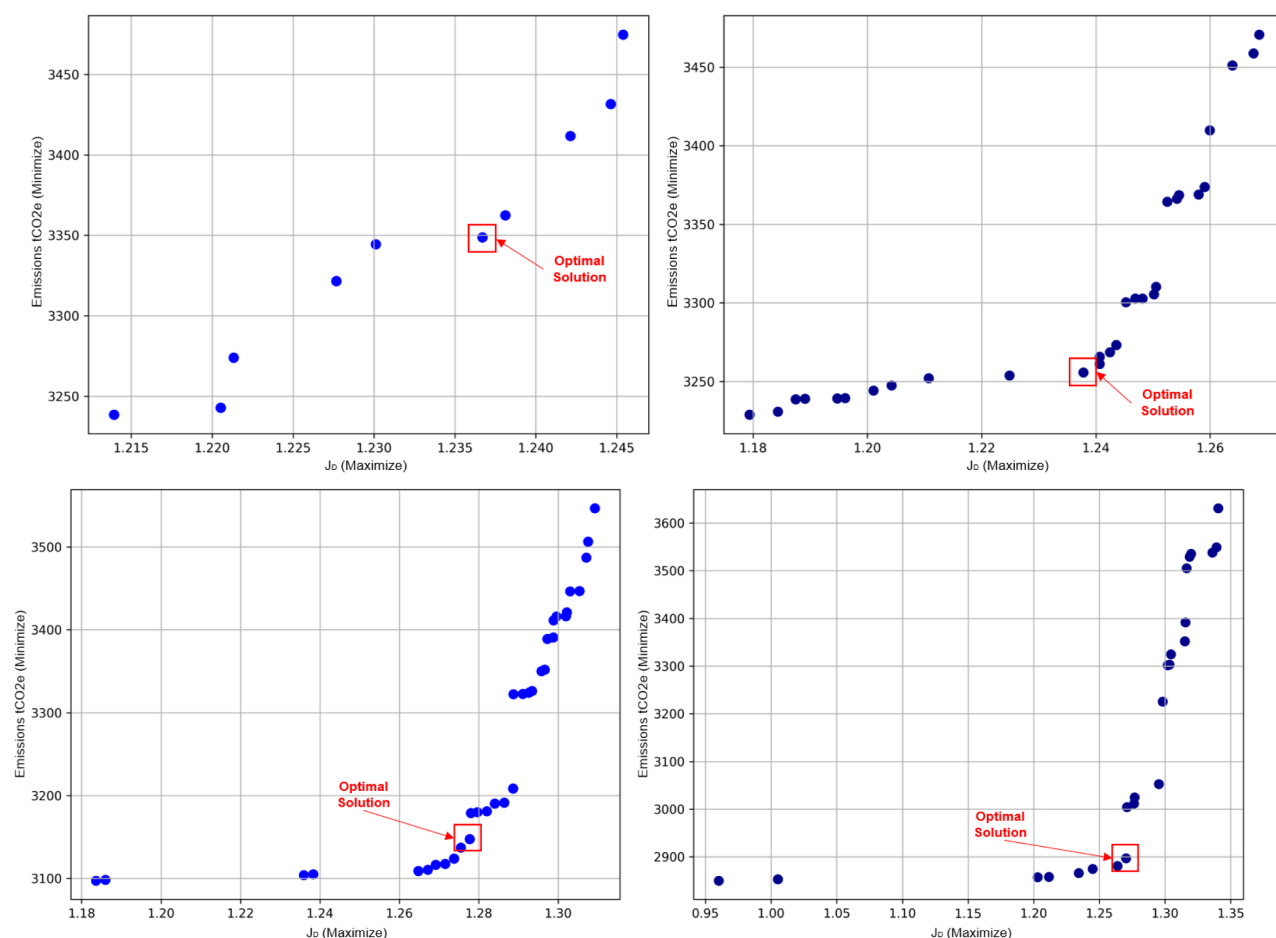


Figure 9—Pareto Front for a single baseline fracture design parameters used for multi-objective optimization to minimize the emissions (tCO₂e) and maximize the dimensionless productivity index (J_D): (top) with range of $\pm 10\%$; (bottom) with range of $\pm 20\%$; (left panel) NSGA-II; (right panel) SMPSO.

Comparing both optimization algorithms, SMPSO's particle-based search mechanism converged more rapidly toward optimal solutions while maintaining a high diversity across solutions. In contrast, NSGA-II required more generations to achieve comparable convergence. In addition, it required a more computationally intensive non-dominated sorting procedure. As a result, SMPSO demonstrated better performance with a wider optimal solution exploration of the trade-offs making it more suitable for this problem. The comparison between the predicted (output from the models) and optimized (output of the optimizer) for both J_D and C can be observed for a constrained range of $\pm 10\%$ and $\pm 20\%$ in Fig. 10. This comparison was done dividing the data into a train set used for model training and a test set used for model evaluation and optimization to assess the optimizer performance and applicability. It can be observed in figure 10 that the $\pm 10\%$ constrained range exhibited a slight decrease of 2.0% on average for J_D. As previously mentioned, this result is expected based on the plateau of J_D with increasing proppant mass, fracture geometry, beyond a point. The tradeoff of 2% reduction in production is a reduction in emissions by 8.2% on average for the entire test set. In addition, we can observe for the J_D results (Figure 10, left), we don't see a significant variation in the upper and lower quartiles, however for the C results (Figure 10, right) there is a noticeable reduction in the upper quartile. If we switch to the results of a wider constrained range of $\pm 20\%$, the degrees of freedom allow the optimizer to expand the search space leading to better results as observed in the figure. On average the J_D values were reduced by 4.1% but generated a reduction in the emissions by 15.8%. Both constrained range cases demonstrated the optimizer potential and the trade-off that can be achieved between increasing production and reducing emissions.

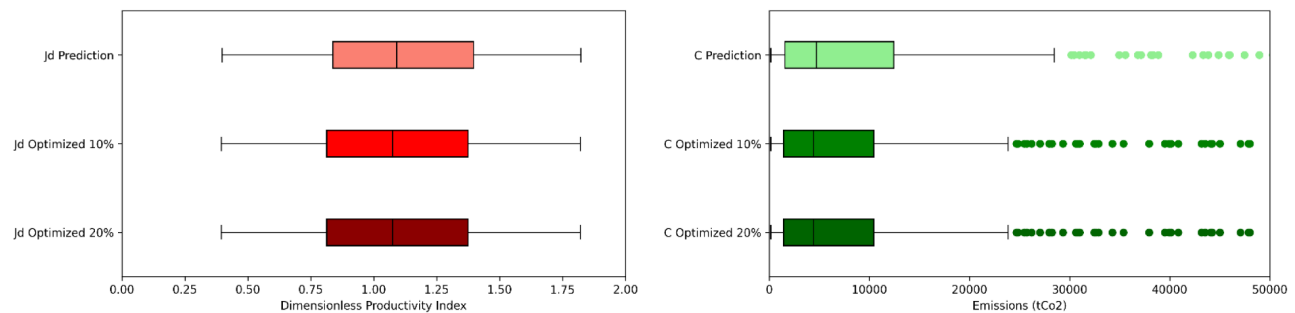


Figure 10—Optimizer results comparing the model prediction and optimized values for the test set; Parameter space varied within $\pm 10\%$ showed a reduction of J_D by 2.0% and a decrease of C by 8.2%; Parameter space varied within $\pm 20\%$ showing a reduction of JD by 4.1% and a decrease of C by 15.8%. (left) J_D ; (right) C.

Finally, after observing the optimizer final results for J_D and C, the distribution of the different fracture design parameters for the constrained range of $\pm 10\%$ can be observed in Fig. 11. Comparing the baseline values against the final optimized values helped understand how variables vary to meet the multi-objective optimization. For fluid efficiency (Eff-XL), the constrained range was only positive, meaning it could only increase. The optimized fracture always prefers higher fluid efficiency on average. From the data analysis, we know proppant mass (M_{prop}) is the predominant feature in both models, plus it is the source for highest emission. As expected, on average there was a reduction of 8.4% in the proppant mass being close to the maximum possible range of 10% reduction. We can observe how the distribution and upper quartiles reduce on the optimized value. For maximum proppant concentration (PPA-Final), and pad ratio (PR_Safe), the average values barely fluctuated with no average difference and 0.2% reduction respectively. However, while the average values remained unchanged compared to the baseline, the maximum values within the optimized results increased, as observed in the boxplot in figure 10. This outcome indicates that although the optimizer did not systematically shift the overall central tendency of the variable, it selectively identified certain cases that were pushing these variables toward its upper bound and contributed to more favorable trade-offs. Essentially, these higher-end values were utilized as some solutions from the test set to achieve overall improvements within the multiple objectives of the optimizer. This behavior shows how multi-objective optimization can uncover targeted adjustments in specific ranges of the search space that could potentially yield to more favorable results.

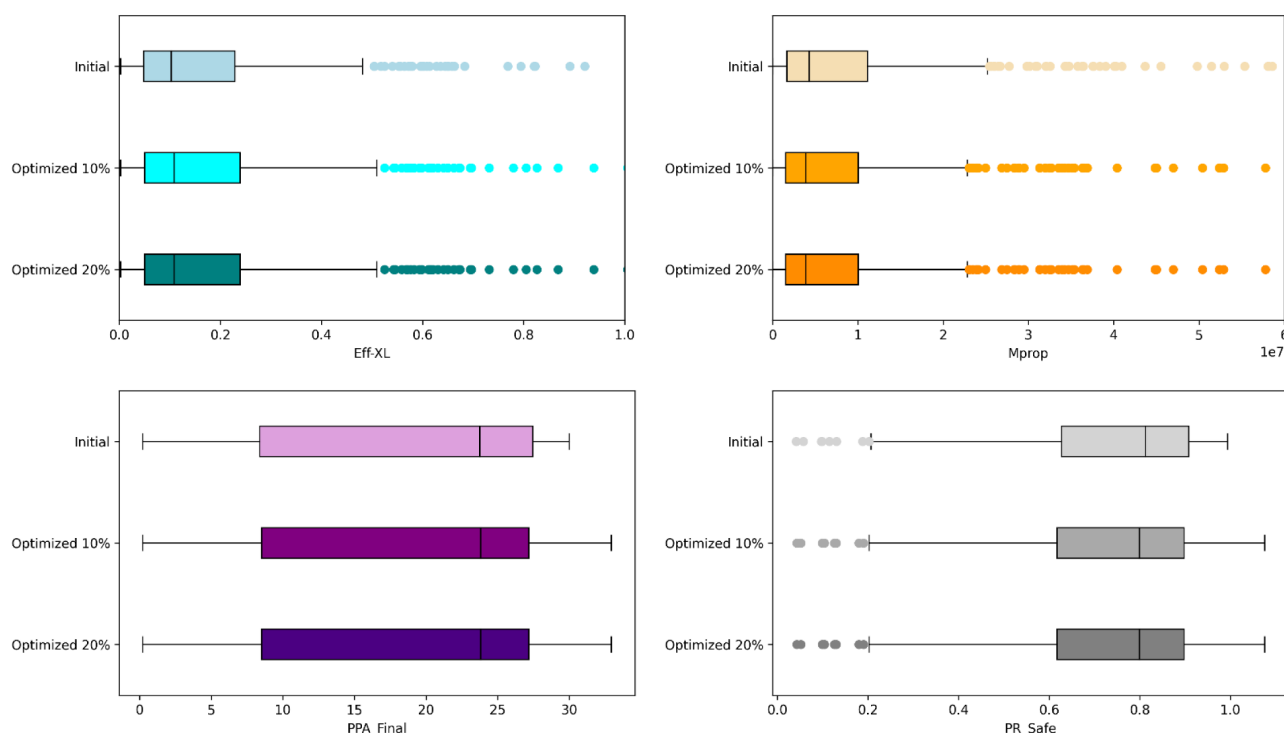


Figure 11—Optimized variables comparison for the test set with range of $\pm 10\%$ and $\pm 20\%$.

Discussion & Way Forward

The dataset construction is internally very consistent which supports high explainability of the model predictions and hence reliability on them. For instance, the treatment choice between conventional or unconventional affects the injection rate choice as well as fluid package, equipment footprint etc. Hence it uses a logic which is sequentially consistent from end-to-end.

The authors acknowledge that even though the introduction builds the conceptual framework on three objective functions, i.e., production, emissions, and economics, the methodology was only implemented for two of these three. The purpose of this research was to postulate the possibility of multivariate optimization and to demonstrate a proof-of-concept. Once this two-factor MOO is achieved the extension to this workflow with additional data for cost, pricing, oil price, etc. inputs is straight forward in order to run the optimization front. It was not considered here to avoid the variability in pricing factors. The authors preferred not to add another layer of high variability, in addition to the existing variability in fracture design and emission parameters.

Another simple option for a well specific workflow is to conduct 10s or possibly 100s of sensitivities (with parallel ensemble) for a well and plot the three objective functions and optimize with simplified derivative and second derivative inflection based logics, somewhat similar to the concept description shown in [figure 1](#). Pareto frontier utilizes a more sophisticated approach and with a larger parameter space which can be generalized and scaled better.

Since emissions intensity is geographically and technologically driven, for the implementation of this prototype, the users will have control over the knobs for various emissions related inputs. For instance, they can change the ranges for embodied carbon, supply chain, and last mile logistics based on their local data.

It is important to mention here that the emissions calculations here do not consider a few critical aspects of the fracturing lifecycle when compared to the detailed LCA modeling conducted by [Khan et al. \(2025a\)](#). The first one is the emissions associated with flaring and venting during post-fracturing flowback. This was not included because of high uncertainty and variance in the amount of fluid recovered during flowing back and hence the amount of gas flared. The second aspect is the emissions associated with the treatment

produced water for reuse. The third aspect is the flowback water transportation to the treatment facility for disposal treatment. Fourth is the emissions with a pipeline water transfer versus trucking. All these factors can be easily added to the dataset but were not considered in this study for simplicity and considering the rarity and complexity of these instances.

Proppant type and mass dominate both production and emissions. Hence recycled or low-carbon ceramics present immediate opportunities for emission reduction and should be considered (Christian et al. 2025). Moreover, the localization of the proppant can significantly affect the supply chain components. Investment and research should be done in this area.

The next steps for this study include the application of this coupled workflow to real-field data to evaluate the extent of benefits that can be extracted from these models. Also, integration and incorporation of these models into real-time digital fracturing advisors could ensure continuous, ESG compliant design and execution. Sustainability metrics can also be extended from emissions to consider water stewardship and toxicity.

Summary and Conclusions

This study introduces a novel PIML framework that integrates sustainability into hydraulic fracturing design by optimizing both production and emissions. Building on the synthetic backbone and transfer learning workflows developed in earlier parts of this series, the dataset was extended to include emissions-related parameters derived from proppant, water, chemical, and equipment factors. Data sources included GREET, GaBi, and EPA LCI datasets, combined with SLB in-house regression models trained on real-time equipment metering.

Two regression models, predicting dimensionless productivity index (J_D) and total carbon emissions (C) were developed using ML regressors. RF and XGBoost yielded best performance metrics achieving R^2 values of 0.7 (J_D) and 0.96 (C). Feature-importance analysis confirmed proppant mass as the dominant variable for both objectives, aligning with domain physics. All the feature ranking was consistent with the causality based on the synthetic dataset construction philosophy. The models were coupled with two optimization architectures: a nested sequential approach (production first, emissions second) and a coupled multi-objective optimization (MOO) approach (simultaneous J_D –C optimization), where the latter was implemented due to its superiority.

Pareto-front analysis demonstrated that the coupled approach consistently delivered superior trade-offs, enabling 8-15% emissions reductions with <2-4% production sacrifice under constrained ± 10 –20% design-variable variation. Among the two tested optimizers, the speed-constrained multi-objective particle swarm optimizer (SMPSO) converged faster and maintained higher solution diversity compared to NSGA-II.

This study demonstrates that sustainable fracturing design can be achieved without significant production penalties. By incorporating carbon intensity as a core design objective, operators can make data-driven decisions balancing economics, production, and ESG targets. The framework is robust, interpretable, and transferable across basins and is well-suited for integration with real-time digital advisory systems.

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Appendix-A

All the variables and parameters utilized for carbon calculations, i.e., the randomization of parameters (Table A1) and the static values (Table A2) along with their references have been detailed in this appendix. The details of the embodied and supply chain associated carbon totals for chemicals and proppant are given in Table A3.

Table A1—Randomized parameters with their type, ranges, and references. It can be observed that the range consideration captures a broad spectrum of jobs, basins, treatment types.

Parameter	Type	Range/ Reference	Unit
Proppant Type	List	Ceramic, white sand, sand	
p_e – Proppant embodied	Randomized from list choice	See Table 3	CO ₂ e-tons / lb
p_{sc} – proppant supply chain	Randomized	See Table 3	CO ₂ e-tons / lb
Proppant, water, equipment source distance	Randomized	0 – 500	km
p_t – proppant last mile	Calculated	NA	CO ₂ e-tons
w – water transportation	Calculated	NA	CO ₂ e-tons
WHP - pressure	Randomized	500 – 15,000	psi
Q – injection rate	Randomized	3 - 120	bbl/min
n – no. of pumps	Calculated	NA	
e_t – equipment last mile	Calculated	NA	CO ₂ e-tons
e – equipment total footprint	Calculated	NA	CO ₂ e-tons
No. of auxiliary equipment	Randomized	5 – 10	
T_{idle} – idling time	Randomized	10 – 60	minutes

Table A2—Static parameters with their references

Parameter	Value	Reference	Unit
Transportation truck fuel efficiency	5.86	US EPA 2005, based on the PERE model	miles/gal
Proppant truck capacity	50,000	Khan et al. 2025a, detailed LCA modeling	lbs
Water transport capacity	6,000	Khan et al. 2025a, detailed LCA modeling	gal
$e_{p,p}$	$= \begin{cases} 45, & Q < 60 \text{ bbl/min} \\ 70, & Q \geq 60 \text{ bbl/min} \end{cases}$	Khan et al. 2025b, regression model from metered data	gal/hr
$e_{p,i}$	8.0	Khan et al. 2025b, regression model from metered data	gal/hr
$e_{blender}$	3.4	Khan et al. 2025b, regression model from metered data	gal/hr
$e_{hydration \text{ unit}}$	3.6	Khan et al. 2025b, regression model from metered data	gal/hr
$e_{auxillary}$	1.0	Khan et al. 2025b, regression model from metered data	gal/hr

Table A3—Embodied and supply chain associated emission components for various chemicals and proppants utilized in the fracturing package. The actual range of values cannot be disclosed due to the rights of the exclusive databases. The table intends to show the breadth of values captured and the sources.

Chemical/Proppant Type	Common Name	Range for p_e and p_{sc}	References for m_e and m_{sc}
Sand	Silica sand / Ottawa sand	not to be disclosed	REET (2022), Ecoinvent v3.8, GaBi, US EPA Smartway (2024)
Ceramics	Sintered bauxite / Kaolin	not to be disclosed	REET, OpenLCA Nexus
Low carbon/ recycled ceramic	-	not to be disclosed	Christian et al. (2025)
Resin coated ceramic	Epoxy coating	not to be disclosed	American Chemistry Council (2022)
Gelling agent	Guar gum	not to be disclosed	REET (2016), Ecoinvent
Crosslinker	Borate or zirconate compounds	not to be disclosed	GaBi, Ecoinvent
Breaker	Ammonium persulfate	not to be disclosed	REET, LCA reports
Surfactant	Nonylphenol ethoxylate	not to be disclosed	Ecoinvent
Biocide	Glutaraldehyde	not to be disclosed	REET
Friction reducer	Polyacrylamide or guar	not to be disclosed	Ecoinvent
Acid	Hydrochloric acid	not to be disclosed	Ecoinvent
Clay stabilizer	KCl or $\text{CH}_4\text{N}_2\text{O} \cdot \text{HCl}$	not to be disclosed	GaBi, REET